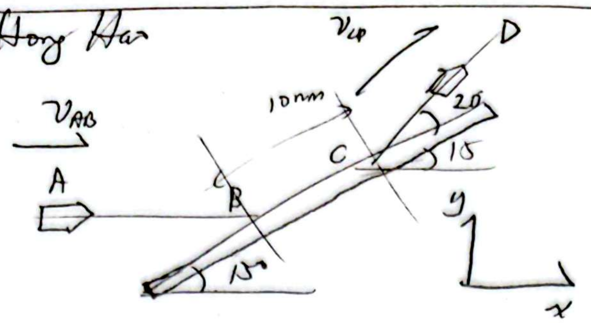
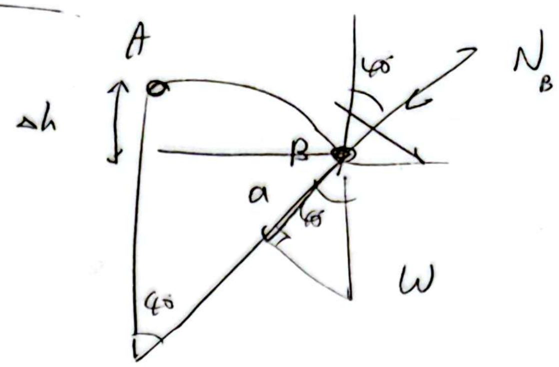
Phang Hong Han





Q 13.195

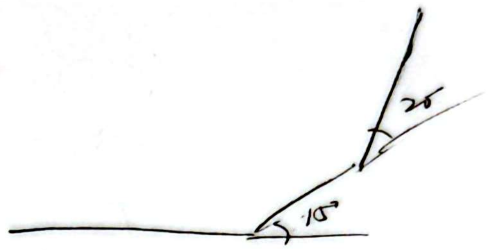
$m = 25 \times 10^{-3}$ kg ,

$|v_{AB}| = 600$ m/s , $|v_{CD}| = 400$ m/s , $|v_{BC}| = 500$ m/s

$(mv)_x = (mv)'_x \; \int_{t_B}^{t_C} F_x dt$

$m(600) = m\frac{(400)}{\cos 35^\circ} + F_x \Delta t$

$\therefore \; F_x \Delta t = I_x = -6.808$ Ns



$\Delta p = F\Delta t = (mv)' - ($

$(mv)_y = (mv)'_y \; \int_{t_B}^{t_C} F_y dt$

$0 = m(400 \sin 35^\circ) \; F_y \Delta t$

$F_y \Delta t = \cancel{5\ Ns} \; 5.736$ Ns

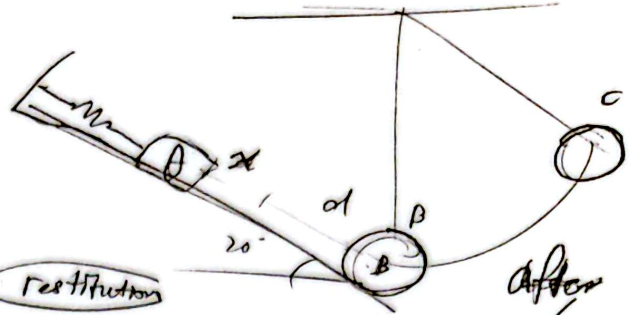
$\because \Delta t = \frac{10 \times 10^{-3}}{500} = 2 \times 10^{-5}$ s

$\therefore \begin{cases} F_x = -3.404 \times 10^5 \text{ N} \\ F_y = 2.868 \times 10^5 \text{ N} \end{cases}$

$F = 4.451 \times 10^5$ N 40.11°

$\alpha = 40°$

13.201 (a) $v_B = ?$ (b) $T_C = ?$



(Collision) $m_A v_{Ax} + 0 = m_A v'_{Ax} + m_B v_B$

restitution

$$e = \frac{v_B - v'_{Ax}}{v_{Ax} - 0} = \frac{v_2 - v_1}{u_1 - u_2}$$ —— ③

After

before

A $\int_0^x kx\,dx - \int_0^{x+d} f\,dx + m_A g h = \frac{1}{2} m_A v_A^2$ —— ①

B $\frac{1}{2} m_B v_B^2 = \frac{1}{2} m_B (v'_B)^2 + mgL(1-\cos\alpha)$ —— ②

①: $\frac{1}{2}kx^2 + m_A g h = \frac{1}{2} m_A v_A^2 + \mu m g \cos\theta (x+d)$

② $\frac{1}{2} m_B v_B^2 = \frac{1}{2} m_B v_B'^2 + mgL(1-\cos\alpha)$

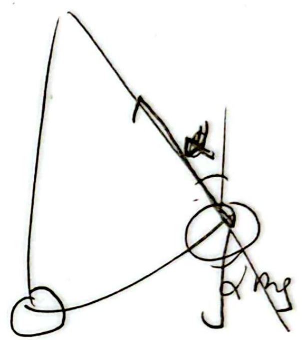
Collision A $\frac{1}{2}(800)(0.1)^2 + (2)(9.8)(x+d)(\sin 20) = \frac{1}{2}(2)(v_A^2) + (0.2)(2)(9.8)\cos 20(1.6)$

B $\frac{1}{2}(1)(v_B^2) = \frac{1}{2}(1)(v_B'^2) + (1)(9.8)(1)(1-\cos 40)$

$0.8 = \frac{v'_B - v'_{Ax}}{v_A - 0}$

3 vars, 3 eqn. Solvable

$\therefore T - m_B g\cos\alpha = \frac{m v^2}{L}$



$\therefore$ (a) $v_B = 2.58$ m/s

(b) $T = 14.15$ N